

A Certified Lower Bound for Packing Ten Unequal Circles in a Circle

Computational proof project

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Abstract

We study the problem of packing ten pairwise non-overlapping circles of radii $\sqrt{1}, \sqrt{2}, \dots, \sqrt{10}$ in a circular container of minimum radius. We present a reproducible interval-arithmetic certificate proving that the optimum radius satisfies

$$7.9 \leq R^* \leq 8.303468122111490.$$

The lower bound uses only a seven-circle subset and all-pair angular inequalities for each of the 360 cyclic orders up to rotation and reversal. A Farkas dual certificate rules out every order, without assuming a contact graph. The upper bound is a conditional feasible-witness bound obtained from a seven-circle interval computation and positive slack checks for the remaining ten-circle constraints. We describe the directed-decimal arithmetic and certificate-replay procedure, including a SciPy-free replay of the angular Farkas cut. The paper deliberately distinguishes implementation-checked computational claims from mathematical implications proved analytically in the text; equality with the displayed upper bound remains an open step of the project.

1 Introduction

Let $r_i = \sqrt{i}$ for $i = 1, \dots, 10$. The unequal-circle packing problem asks for the smallest R for which there exist centers $p_i \in \mathbb{R}^2$ such that

$$\|p_i\| \leq R - r_i, \quad i = 1, \dots, 10, \quad (1)$$

$$\|p_i - p_j\| \geq r_i + r_j, \quad 1 \leq i < j \leq 10. \quad (2)$$

We write R^* for this minimum.

The minimum exists: for any bounded interval of container radii the feasible set is closed, and a feasible configuration is contained in a compact disk. The area bound gives a finite lower bound, while any sufficiently large container is feasible, so the infimum is attained.

The main contribution of this paper is a lower-bound argument that does not depend on a guessed optimal contact graph. The key observation is that a finite set of centers surrounding the origin must spend a total angular budget of 2π . Pairwise non-overlap gives lower bounds for angular path lengths. For each possible cyclic order, these inequalities are combined by a Farkas dual certificate.

Our result is intentionally modest. It does not prove that the currently known feasible configuration is globally optimal. Instead, it establishes a strong, reproducible lower bound for subsequent global exclusion arguments.

2 Problem formulation

Since the area bound gives $R \geq \sqrt{55} > \sqrt{10}$, it suffices to consider $\rho \geq \sqrt{10}$. For such ρ , feasibility is equivalently expressed by

$$\|p_i\|^2 \leq (\rho - r_i)^2, \quad (3)$$

$$\|p_i - p_j\|^2 \geq (r_i + r_j)^2. \quad (4)$$

The area bound gives $R \geq \sqrt{55}$, but is substantially weaker than the bound obtained below. Since every feasible ten-circle configuration restricts to every subset of its circles, infeasibility of a subset at radius ρ implies infeasibility of the full problem at the same radius.

3 The angular-cycle lower bound

Fix a subset S of circles. Put $a_i = \|p_i\|$. Containment gives $a_i \leq \rho - r_i$. For any $j \in S \setminus \{i\}$, the triangle inequality and non-overlap give

$$a_i \geq r_i + r_j - a_j \geq r_i + 2r_j - \rho.$$

Consequently every feasible configuration satisfies

$$L_i(\rho) := \max \left(0, \max_{j \in S \setminus \{i\}} (r_i + 2r_j - \rho) \right) \leq a_i \leq U_i(\rho) := \rho - r_i. \quad (5)$$

For the certificate below, $L_i(\rho) > 0$ for every $i \in S$, so all selected center vectors are nonzero. For two nonzero centers define θ_{ij} to be their smaller central angle. The law of cosines gives

$$(r_i + r_j)^2 \leq a_i^2 + a_j^2 - 2a_i a_j \cos \theta_{ij}.$$

Hence the required angular lower bound is defined by

$$\alpha_{ij}(\rho) := \min_{\substack{a_i \in [L_i(\rho), U_i(\rho)] \\ a_j \in [L_j(\rho), U_j(\rho)]}} \arccos \left(\frac{a_i^2 + a_j^2 - (r_i + r_j)^2}{2a_i a_j} \right). \quad (6)$$

Equivalently, if

$$c_{ij}^{\max}(\rho) := \max_{\substack{a_i \in [L_i(\rho), U_i(\rho)] \\ a_j \in [L_j(\rho), U_j(\rho)]}} \frac{a_i^2 + a_j^2 - (r_i + r_j)^2}{2a_i a_j},$$

then $\alpha_{ij}(\rho) = \arccos c_{ij}^{\max}(\rho)$, with the usual endpoint convention when the enclosing cosine bound lies outside $[-1, 1]$.

The maximization of the cosine expression over that rectangle may be reduced to the four corners. Indeed, for fixed b its derivative with respect to a has at most one zero, and that stationary point is a minimum, so the maximum in a occurs at an endpoint. Evaluating those two endpoints and then applying the same argument to b leaves the four corners. We evaluate the four corner values with outward-rounded decimal arithmetic and then apply the decreasing function $\arccos$ using a directed enclosure.

Let $w_{ij} = \alpha_{ij}(\rho)$. If the centers in S are listed in their cyclic order around the origin, the sum of the consecutive angular gaps is 2π . Therefore a necessary condition for feasibility is

$$\min_{C \text{ Hamiltonian cycle on } S} \sum_{(i,j) \in C} w_{ij} \leq 2\pi. \quad (7)$$

The minimum is computed by the Held–Karp dynamic program. No cyclic order is assumed in advance.

Proposition 3.1 (Angular-cycle infeasibility criterion). *If the left-hand side of (7) is strictly larger than 2π , then the subset S cannot be packed in a container of radius ρ .*

Proof. Every feasible realization has a cyclic ordering of its nonzero center vectors. Each consecutive gap is bounded below by the corresponding w_{ij} , and the gaps sum to 2π . Its cyclic ordering is one of the Hamiltonian cycles considered in (7), contradicting the strict inequality. The radial lower bounds ensure that the selected centers are nonzero in the certificate used below. $\square$

For a stronger certificate, fix one cyclic order and write its seven angular gaps as $x_0, \dots, x_6$. For every pair of positions there are two directed paths around the cycle. If $q_k \in \{0, 1\}^6$ is the incidence vector of one such path and b_k is the corresponding pair-angle lower bound, feasibility requires

$$q_k \cdot x \geq b_k \quad (k = 1, \dots, 42), \quad x \geq 0, \quad \sum_g x_g = 2\pi. \quad (8)$$

Proposition 3.2 (Fixed-order Farkas certificate). *Suppose $y_k \geq 0$ satisfies*

$$\sum_{k=1}^{42} y_k q_k[g] \leq 1 \quad (g = 0, \dots, 6) \quad (9)$$

and $\sum_k y_k b_k > 2\pi$. Then the selected cyclic order is impossible.

Proof. Multiplying the inequalities in (8) by y_k and summing, then using (9), gives

$$\sum_k y_k b_k \leq \sum_k y_k (q_k \cdot x) \leq \sum_g x_g = 2\pi,$$

which contradicts the assumed strict inequality. $\square$

4 Certified computation

All interval endpoints are finite decimal numbers. Arithmetic operations are rounded toward the appropriate direction. The numerical data are supplied as machine-readable supplementary certificates. The replay verifier reconstructs the model, recomputes the directed bounds, and checks the recorded inequalities; the logical implication from those inequalities to infeasibility is proved in the preceding propositions.

For the present result we use

$$S = \{4, 5, 6, 7, 8, 9, 10\}, \quad \rho = 7.9.$$

There are $(7 - 1)!/2 = 360$ cyclic orders after identifying orders related by rotation and reversal. Reversal need not be considered separately because both directed paths for every pair are included. For each of these 360 cyclic orders, the radial interval for each circle is split into two subintervals. The resulting $360 \cdot 2^7 = 46080$ radial boxes each have a certificate with nonnegative Farkas multipliers for all 42 directed path inequalities. The MPFI replay verifies that every resulting dual value strictly exceeds the upward-rounded bound $2\pi_{\text{upper}}$, where

$$2\pi_{\text{upper}} = 6.283185307179586476925286766559 \dots$$

The accompanying machine-readable certificate contains outward-rounded box endpoints and decimal multipliers for all 46080 boxes. The MPFI replay checks all dual column inequalities and all directed angle bounds; SciPy is used only to discover the multipliers. The analytic argument above is self-contained, while the displayed numerical conclusion is conditional on the supplementary certificate and its MPFI implementation.

Theorem 4.1 (Certified lower bound). *Subject to the stated directed-decimal model and replay of the supplementary certificate,*

$$R^* \geq 7.9.$$

Proof. The seven-circle subset S is infeasible at $\rho = 7.9$ by the preceding certificate. Any ten-circle packing would restrict to a packing of S . Therefore the ten-circle problem is infeasible at that radius, and the optimal radius is at least 7.9. $\square$

5 The feasible upper bound

A Krawczyk-certified active-core root has radius coordinate in the interval

$$8.3034681221114890787043811875161 \leq R_{\text{root}} \leq 8.3034681221114890787043811875163. \quad (10)$$

Here $R_{\text{root}} = \hat{R}_{\text{root}}$ is the radius coordinate of the Krawczyk root. We use the finite decimal

$$\overline{R} = 8.303468122111490$$

as the certified container-radius upper bound. Its active seven-circle core consists of circles $\{2, 5, 6, 7, 8, 9, 10\}$. The active wall contacts are $\{2, 5, 6, 7, 8, 9\}$ and the active pair contacts are

$$(5, 7), (2, 8), (6, 8), (2, 9), (7, 9), (5, 10), (6, 10), (7, 10).$$

To remove the rotational degree of freedom, we impose $y_{10} = 0$ and take $x_{10} > 0$ in the displayed box. The Krawczyk unknown vector is

$$z = (x_2, y_2, x_5, y_5, x_6, y_6, x_7, y_7, x_8, y_8, x_9, y_9, x_{10}, \hat{R}),$$

so it has 14 variables. The square system has the six equations

$$\|p_i\|^2 - (\hat{R} - r_i)^2 = 0 \quad (i = 2, 5, 6, 7, 8, 9)$$

and the eight equations

$$\|p_i - p_j\|^2 - (r_i + r_j)^2 = 0$$

for the eight listed active pairs. A Krawczyk inclusion on the symmetric box of half-width 10^{-65} in these 14 variables encloses a unique root. The remaining circles 1, 3, 4 are placed at the listed decimal coordinates and are checked using the corresponding interval enclosure. All 41 inactive wall and pair inequalities have strictly positive interval slack in the squared forms of the constraints (3)–(4); the smallest reported lower bound is approximately 0.0404383472086741 for the wall constraint of circle 1.

For reproducibility, the following table gives a decimal presentation of the realization used for the upper-bound check. The coordinates of the seven active-core circles are rounded from the corresponding interval enclosure; the three rattler coordinates are the fixed decimal coordinates used in the extension check. The radius of circle i is $r_i = \sqrt{i}$.

Table 1: Centers of the certified upper-bound realization.

i	x_i	y_i	r_i
1	-3.979411598680	-6.120824420924	$\sqrt{1}$
2	-6.643004688846	1.825463528216	$\sqrt{2}$
3	6.491367788756	-0.773581798891	$\sqrt{3}$
4	5.592388025100	2.873904150833	$\sqrt{4}$
5	4.323261823214	-4.257082536523	$\sqrt{5}$
6	1.885281861457	5.542091227010	$\sqrt{6}$
7	-0.356672093417	-5.646463010751	$\sqrt{7}$
8	-3.266260803898	4.394043045277	$\sqrt{8}$
9	-4.826665378217	-2.197743262753	$\sqrt{9}$
10	1.003716086075	0.000000000000	$\sqrt{10}$

The displayed decimals are for identification and reproduction only; the feasibility assertion uses the enclosing intervals rather than these rounded values. In particular, the active equalities are checked on the interval core, while all inactive inequalities are checked with strictly positive lower slack.

Figure 1 visualizes the same realization. The TikZ coordinates are the displayed center coordinates in Table 1; the common scale changes only the size of the drawing, not the geometry.

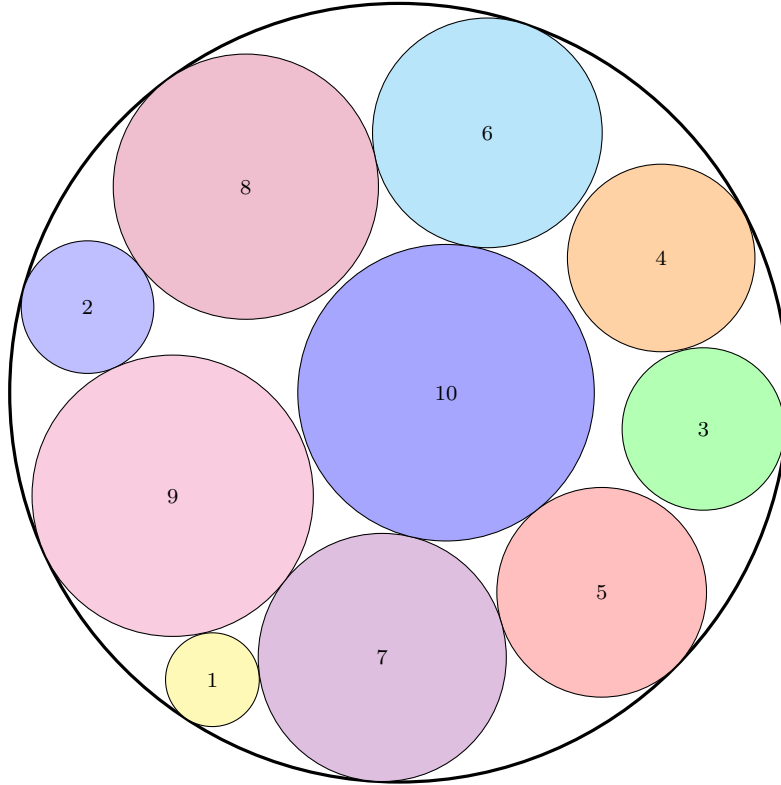


Figure 1: Visualization of the certified upper-bound realization. The plotted centers use the rounded coordinates in Table 1; the outer circle has radius $\bar{R}$.

Thus, conditional on the supplementary interval enclosure and Krawczyk certificate,

the configuration gives

$$R^* \leq \overline{R}. \tag{11}$$

Because containment is monotone in the container radius, a configuration feasible at R_{root} is also feasible at the larger finite-decimal radius $\overline{R}$. This is a feasible-witness statement, not an optimality claim.

6 Conclusion and open problems

Combining the certified lower bound with the feasible witness gives

$$\boxed{7.9 \leq R^* \leq 8.303468122111490.}$$

The next proof-critical tasks are a global exclusion for every radius below $\overline{R}$, complete treatment of minimal positive stress supports including degenerate cases, and rigorous extension tests for circles not contained in a candidate stressed support. The current result does not assert $R^* = \overline{R}$; the displayed upper bound is not claimed to be optimal.